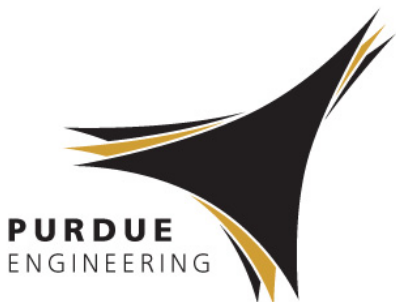


Spring, 2015

ME 612 – Continuum Mechanics

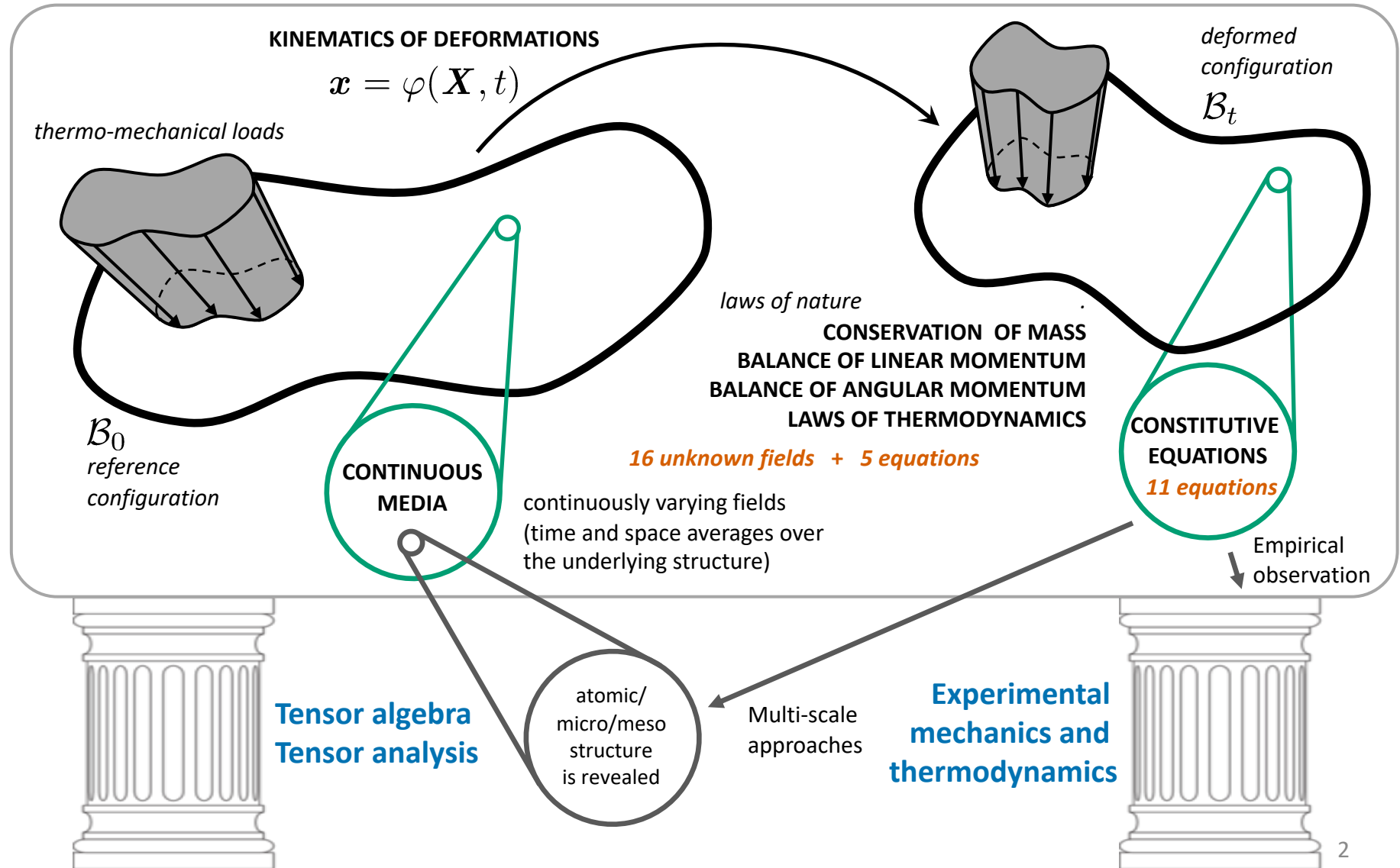
Lecture 2

Introduction to vectors and tensors



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Lecture 2 – Introduction to tensors and vectors



Lecture 2 – Introduction to tensors and vectors

Review:

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i =$$

$$\delta_{ii} =$$

$$\epsilon_{ijk} \epsilon_{ijk} =$$

$$\epsilon_{ijk} \delta_{ij} =$$

DIY

What is a tensor?

- Tensors are abstract mathematical entities.
- Vectors are *first order* tensors.
- Vectors and tensors exist separately of a particular coordinate system (i.e., they are coordinate invariant).

... so let's start by reviewing Vector Algebra.

Vector algebra

Vector space

A real vector space V is a set, defined over the field of real numbers $\mathbb{R}$, where the following two operations have been defined:

- vector addition: for $\mathbf{a}, \mathbf{b} \in V$, then $\mathbf{a} + \mathbf{b} = \mathbf{c} \in V$
- scalar multiplication: for $\lambda \in \mathbb{R}$ and $\mathbf{a} \in V$, then $\lambda \mathbf{a} = \mathbf{c} \in V$

with the following properties $\forall \mathbf{a}, \mathbf{b}, \mathbf{c} \in V$ and $\forall \lambda, \mu \in \mathbb{R}$:

1. $\mathbf{a} + \mathbf{c} = \mathbf{b} + \mathbf{a}$
2. $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$
3. $\mathbf{a} + \mathbf{0} = \mathbf{a}$
4. $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$
5. $\lambda \mathbf{a} = \mathbf{a} \lambda$
6. $\lambda(\mu \mathbf{a}) = (\lambda \mu) \mathbf{a}$
7. $1 \mathbf{a} = \mathbf{a}$
8. $(\lambda + \mu)(\mathbf{a} + \mathbf{b}) = \lambda \mathbf{a} + \lambda \mathbf{b} + \mu \mathbf{a} + \mu \mathbf{b}$

DIY

Which is the name of each property?

Vector algebra

Vector space – Euclidean Space

- In continuum mechanics, we restrict attention to finite-dimensional spaces.
- We also need additional geometric properties, such as distances and angles.



Euclidean
Space

... but ...

- What is an n -dimensional vector space?
- What is a distance?
- What is an angle?

Naturally, you have a good intuitive understanding of these concepts, but we will formalize them with the purpose of understanding their generalization.

What is an n-dimensional vector space?

- Linear independent vectors

Given $(\mathbf{e}_1, \dots, \mathbf{e}_{n_d})$,

$$\lambda_i \mathbf{e}_i = \mathbf{0} \text{ iff } \lambda_i = 0,$$

then $(\mathbf{e}_1, \dots, \mathbf{e}_{n_d})$ are linearly independent

- Given a vector space, the largest possible number of linearly-independent vectors is the dimensionality of the vector space.

$$\dim V = n_d \quad V \equiv V^{n_d}$$

- Any set of n_d linearly independent vectors can be selected as a basis of V^{n_d} .
- Every vector can be written as a unique linear combination of the basis vectors.

Given $(\mathbf{e}_1, \dots, \mathbf{e}_{n_d})$, $\forall \mathbf{a} \in V^{n_d}, \exists$ unique a_i s.t. $\mathbf{a} = a_i \mathbf{e}_i$

What is a distance? What is an angle?

- Both can be defined from geometric interpretations of an inner product and a norm.

The inner product of two vectors $\mathbf{a}$ and $\mathbf{b}$ is a real-valued bilinear mapping denoted by $\langle \mathbf{a}, \mathbf{b} \rangle$. It must satisfy the following properties $\forall \mathbf{a}, \mathbf{b}, \mathbf{c} \in V$ and $\forall \lambda, \mu \in \mathbb{R}$:

1. $\langle \lambda \mathbf{a} + \mu \mathbf{b}, \mathbf{c} \rangle = \lambda \langle \mathbf{a}, \mathbf{c} \rangle + \mu \langle \mathbf{b}, \mathbf{c} \rangle$
2. $\langle \mathbf{a}, \mathbf{b} \rangle = \langle \mathbf{b}, \mathbf{a} \rangle$
3. $\langle \mathbf{a}, \mathbf{a} \rangle \geq 0$ and $\langle \mathbf{a}, \mathbf{a} \rangle = 0$ iff $\mathbf{a} = \mathbf{0}$

Given an inner product, the associated norm of a vector $\mathbf{a} \in V$ is defined as

$$\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle}$$

Moreover, every inner product satisfies the Cauchy-Schwartz inequality for all $\mathbf{a}, \mathbf{b} \in V$

$$|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

and therefore we can define θ as

$$\cos \theta = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

Vector algebra

What is a distance? What is an angle?

- Both can be defined from geometric interpretations of an inner product and a norm.
- Euclidean space:

Euclidean
Space

$$V = \mathbb{R}^{n_d}$$

$$\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a} \cdot \mathbf{b}$$

$\|\mathbf{a}\|$ provides a measure of the length of the vector $\mathbf{a}$

$\theta(\mathbf{a}, \mathbf{b})$ provides the angle between vectors $\mathbf{a}$ and $\mathbf{b}$

DIY

Let's recall addition and multiplication in $\mathbb{R}^{n_d}$

Vector algebra

What is a distance? What is an angle?

- Both can be defined from geometric interpretations of an inner product and a norm.
- Euclidean space $\mathbb{R}^{n_d}$

Euclidean
Space

- Euclidean point space E^{n_d}

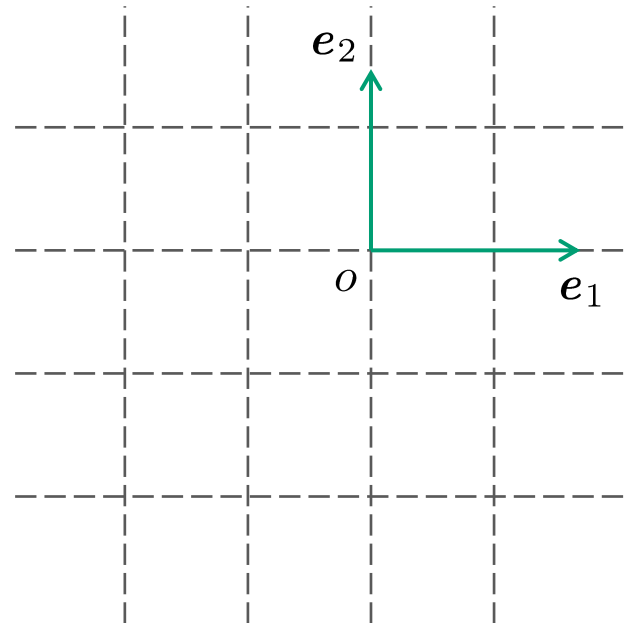
A set of points $x \in E^{n_d}$ uniquely associated with position vectors $\mathbf{x} \in \mathbb{R}^{n_d}$ defined with respect to an arbitrary origin or reference point o .

- Now we can naturally extend the notion of distance and angle to:
 - + Distance between two points
 - + Angle formed by three points

Vector algebra

Coordinate systems

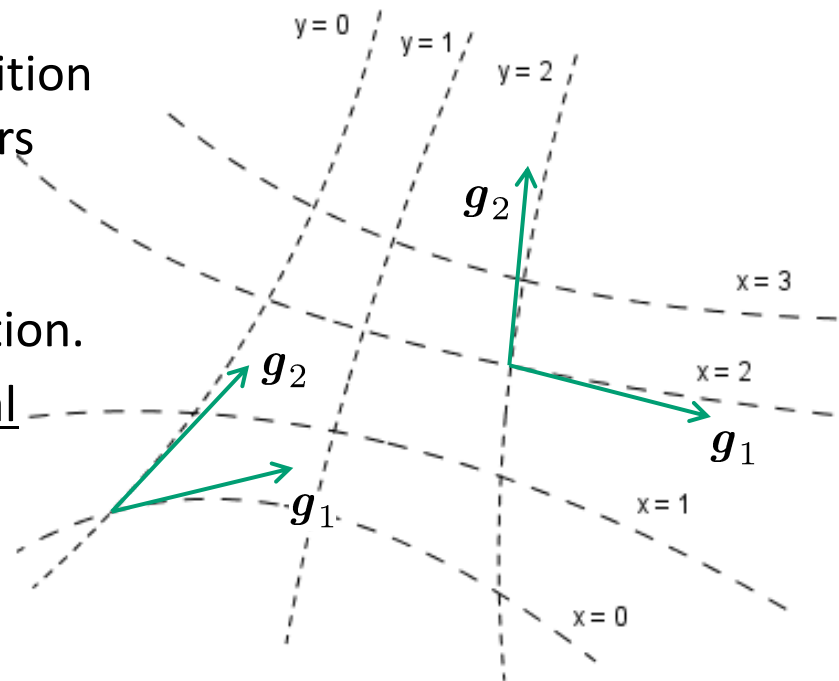
- An origin (relative to which positions are measured)
- A set of coordinate curves (that correspond to paths through space along which all but one of the coordinates are constant)



Vector algebra

Coordinate systems

- An origin (relative to which positions are measured)
- A set of coordinate curves (that correspond to paths through space along which all but one of the coordinates are constant)
- Basis are defined at each position in space as the tangent vectors to the coordinate curves. Therefore, basis vectors $\{g_i\}$ change from position to position.
- Basis $\{g_i\}$ are non-orthogonal in general.



Coordinate systems

- If the coordinate curves are straight lines then the system is a rectilinear coordinate system.
- If the basis vectors of a rectilinear coordinate system are orthogonal then the system is a Cartesian coordinate system.
- If the basis vectors (or axes) of a Cartesian coordinate system are unit vectors $\{e_i\}$ then the basis are called orthonormal and follow the condition

$$e_i \cdot e_j = \delta_{ij}$$

- We will choose basis vectors that form a right-handed triad

$$e_i \times e_j = \epsilon_{ijk} e_k$$

What is the component of a vector (or the coordinate of a position vector) along a basis vector direction?

DIY

Vector algebra

Curvilinear coordinate systems

- Two set of basis vectors at each position in space

$\{\mathbf{g}_i\}$ tangent vectors

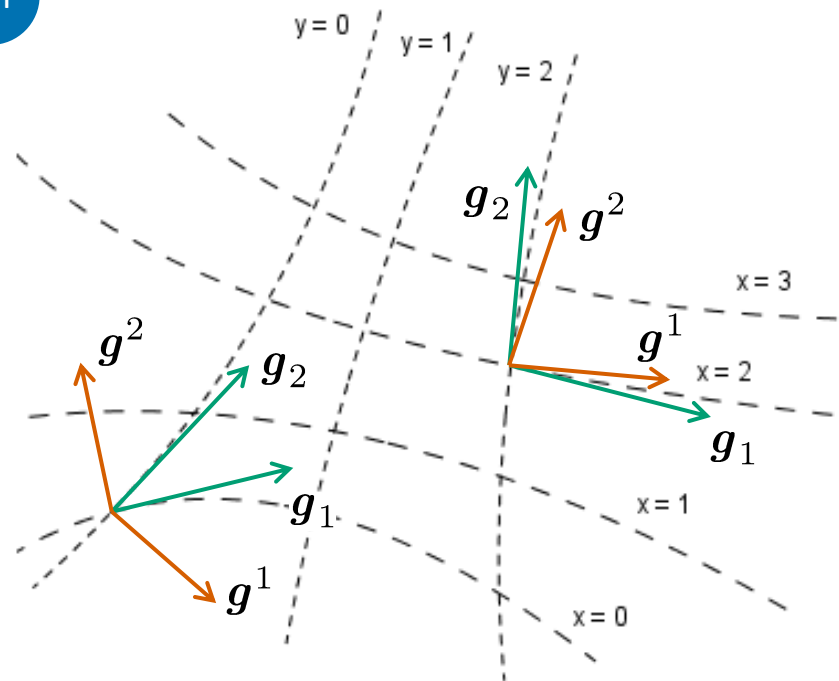
$\{\mathbf{g}^i\}$ reciprocal vectors

defined through $\mathbf{g}^i \cdot \mathbf{g}_j = \delta_j^i$

DIY

What is the component of a vector along a basis vector direction?

- Examples:
Polar cylindrical coordinates
Spherical coordinates



Vector algebra

Curvilinear coordinate systems

- Two set of basis vectors at each position in space

$\{\mathbf{g}_i\}$ tangent vectors

$\{\mathbf{g}^i\}$ reciprocal vectors

defined through $\mathbf{g}^i \cdot \mathbf{g}_j = \delta_j^i$

- Contravariant components

$$\mathbf{a} = a^i \mathbf{g}_i$$

- Covariant components

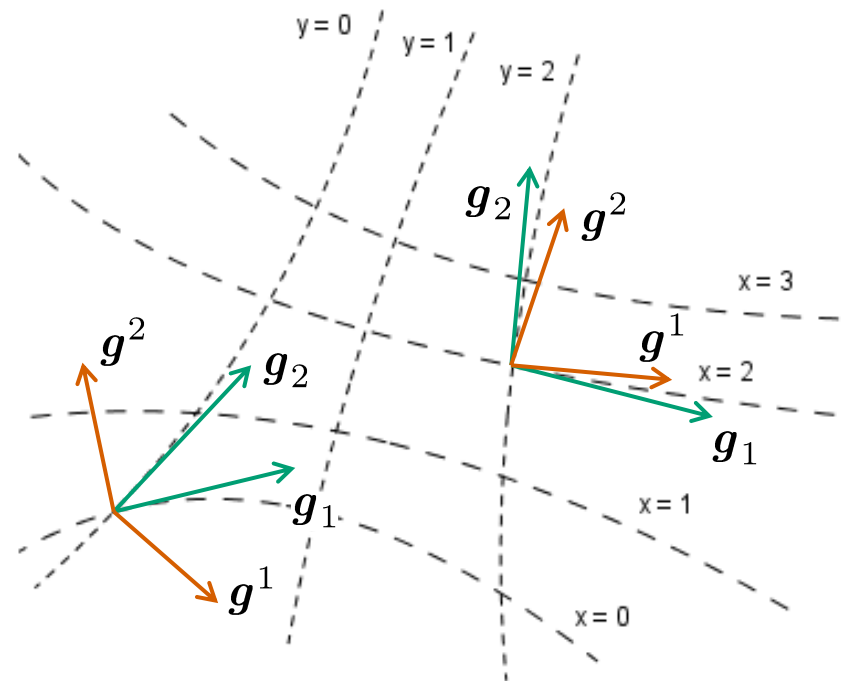
$$\mathbf{a} = a_j \mathbf{g}^j$$

- Connection between components

$$a^k = g^{jk} a_j$$

$$a_k = g_{ik} a^i$$

with $g^{jk} = \mathbf{g}^j \cdot \mathbf{g}^k$
 $g_{ik} = \mathbf{g}_i \cdot \mathbf{g}_k$



Vector algebra

Curvilinear coordinate systems

- All vector and tensor related operations (and continuum mechanics in general) can be defined in curvilinear coordinate systems
- Example: the dot product

$$\mathbf{a} \cdot \mathbf{b} = (a^i \mathbf{g}_i) \cdot (b^j \mathbf{g}_j) = a^i a^j (\mathbf{g}_i \cdot \mathbf{g}_j) = a^i a^j g_{ij}$$

$$\mathbf{a} \cdot \mathbf{b} = a_i b_j g^{ij}$$

$$\mathbf{a} \cdot \mathbf{b} = a_i b^i$$

$$\mathbf{a} \cdot \mathbf{b} = a^i b_i$$

Note: g^{jk} and g_{ik} are components of the metric tensor.

We will *mostly* limit ourselves to Cartesian coordinate systems

Change of basis (... back to Cartesian systems)

- Two orthonormal bases $\{e_\alpha\}$ and $\{e'_i\}$
- The goal is to change bases from $\{e_\alpha\}$ to $\{e'_i\}$
- A linear transformation matrix $\mathbf{Q}$ is defined as

$$e'_i = Q_{\alpha i} e_\alpha \quad Q_{\alpha i} = e_\alpha \cdot e'_i$$

DIY

Let's write the transformation using a column-matrix notation.

Show the following properties:

$$\mathbf{Q}^T = \mathbf{Q}^{-1}$$

$$\det(\mathbf{Q}) = \pm 1$$

Vector algebra

Vector component transformation

- We opened this Lecture by saying that
Vectors and tensors are coordinate invariant, i.e., invariant with respect to component transformation.

$$\mathbf{a} = a_{\alpha} \mathbf{e}_{\alpha} = a'_i \mathbf{e}'_i = a'_i (Q_{\alpha i} \mathbf{e}_{\alpha})$$

Prove the following transformation rules

$$a_{\alpha} = Q_{\alpha i} a'_i$$

$$a'_i = Q_{\alpha i} a_{\alpha}$$

DIY

Vector component transformation

- We opened this Lecture by saying that
Vectors and tensors are coordinate invariant, i.e., invariant with respect to component transformation.

$$\mathbf{a} = a_\alpha \mathbf{e}_\alpha = a'_i \mathbf{e}'_i = a'_i (Q_{\alpha i} \mathbf{e}_\alpha)$$

- A vector $\mathbf{a} \in \mathbb{R}^n$ can be expressed as components a_i in $\{\mathbf{e}_i\}$.

Alternatively, it can be replaced by a linear mapping

$$\mathbf{a}[\mathbf{d}] = \mathbf{a} \cdot \mathbf{d}$$

which returns a real number equal to the projection of the vector on $\mathbf{d}$. Therefore, when evaluated on orthonormal Cartesian basis, it returns the coordinates of the vector

$$a_i \equiv \mathbf{a}[\mathbf{e}_i]$$

What is a tensor?

From first order to n -th order tensors

- A first order tensor (vector) is a real-valued linear function of vectors

$$\mathbf{a} : \mathbb{R}^{n_d} \rightarrow \mathbb{R} \qquad a_i \equiv \mathbf{a}[\mathbf{e}_i]$$

- An n -th order tensor is a real-valued n -linear function of vectors

$$\mathbf{A} : \mathbb{R}^{n_d} \times \dots \times \mathbb{R}^{n_d} \rightarrow \mathbb{R}$$

- The components of a second-order tensor in a particular basis $\{\mathbf{e}_i\}$ are then defined as

$$A_{ij} \equiv \mathbf{A}[\mathbf{e}_i, \mathbf{e}_j]$$

$$\mathbf{T}[\mathbf{a}, \mathbf{b}] =$$

DIY

Lecture 2 – Introduction to tensors and vectors

Any questions?